When Sample Size Misprices Sampling Cost: A Decision-Impact Analysis of Configurable-System Performance Prediction

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***Abstract*—The number of measured configurations is the standard effort axis in sampling-based performance prediction. We ask when that axis is misleading. Across six evidence families—revisiting published configurable-system studies and extending to performance-energy, workload sensitivity, hyperparameter-learning curves, and Linux kernel releases—we replace sample count with measured acquisition cost and track what changes.**

The answer: sample size orders effort but does not budget it. Same-size samples can differ by 16.5× in measured cost. Replacing count by measured cost changes the recommended strategy-budget combination in 538/1,666 decisions (32%), and in 57 cases the count winner is strictly dominated under measured cost. Pareto fronts miss about 52% of measured-cost-efficient combinations. The problem is strongest under heavy-tailed costs, target-cost asymmetry, and diversity-seeking samplers.

The practical fix is a lightweight three-step audit: inspect per-configuration cost spread, report per-strategy budget ratios, and recompute rankings and Pareto fronts under measured cost. We show how this audit changes conclusions across six evidence families and provide artifact support for future studies to run it themselves.

***Index Terms*—configurable systems, performance prediction, sampling, measurement cost, empirical software engineering**

I. Introduction

When should a practitioner trust sample size as a proxy for sampling cost? Figure 1 shows why the answer matters. On lrzip, five sampling strategies each measure the same 16 configurations. Under the standard count-based view they are equally expensive. Under measured acquisition cost—the sum of the runtimes actually spent—pilot-cost sampling spends 0.034% of the full measurement campaign while diversity sampling spends 0.567%. Their prediction errors are nearly identical (51.3% vs. 52.4% MRE), yet the same-size samples differ by $16.5\times$ in real cost. Sample size is a count, but it may not be a budget.

How often does this happen? Across six evidence families and $1{,}666$ strategy–budget decisions, replacing count by measured cost changes the recommended strategy–budget combination in 538 cases (32%). In 57 of those the count winner is strictly dominated—another combination is no less accurate and no more expensive under measured cost. Count-based Pareto fronts miss about 52% of the combinations that are efficient under measured cost. The proxy failure is not rare.

This paper's thesis is deliberately actionable:

> *Practitioners should not trust sample size as a cost proxy without running a lightweight three-step audit: (1) inspect per-configuration cost spread, (2) report per-strategy budget ratios, and (3) recompute rankings and Pareto fronts under measured cost.*

The audit asks three questions. First, are per-configuration costs homogeneous enough that a count is a reasonable budget? (Heavy tails and orders-of-magnitude spreads are warning signs.) Second, do strategies select systematically cheaper or costlier configurations, so that a same-size comparison hides a real cost difference? (The budget ratio $Ratio$ exposes this.) Third, does the ranking or efficient set change when measured cost replaces sample count? (If yes, the count proxy is not valid for that decision.)

We test the audit across six complementary evidence families. The first two are replications of published configurable-system studies: a REPLICATION-ICSE19 revisit of Kaltenecker-style systems and a REPLICATION-ICPE20 revisit of x264 over 17 inputs and two properties. The remaining four stress the audit under new conditions: a performance–energy corpus (ENERGY), a workload-impact corpus with 93 system–workload contexts (WORKLOAD), an external hyperparameter-optimization benchmark (LCBENCH), and seven Linux kernel releases with storage-bound cost (LINUX). These families are stress tests: if the audit validates the count proxy in some and flags it in others, it shows the check is context-dependent and necessary.

Figure 2 makes the same point in a stronger form: a `pilotCost` sample of 256 configurations is both cheaper and slightly more accurate than a `diversity` sample of 128. Under a count budget the larger sample is simply "more expensive"; under measured cost it is the cheaper, better choice. This is not an isolated artifact—it is the same mechanism that drives the $538/1{,}666$ decision changes across the full study.

The problem is not that sample size is uncorrelated with cost. It is that correlation is not enough for decision-making. A practitioner comparing two strategies at a fixed n needs to know whether they spend similar measured budgets. A reader of a sampling paper needs to know whether a claim such as "strategy A is more cost-effective than B" means fewer configurations or less measured resource. These are different statements, and our audit makes them distinguishable.

Contributions.

- A lightweight three-step audit for checking whether sample size is a valid cost proxy in configurable-system sampling studies.

Sampling principle. A strategy selects n configurations from the option space; the measured runtime of those configurations is the acquisition cost, and their energy labels train the predictor.

$$\mathcal{C} \xrightarrow[\text{strategy } s]{n=16} S_{s,n}, \qquad TrainCost = \sum_{c \in S_{s,n}} \text{runtime}(c)$$

Excerpt of selected rows.

Strategy	enc.	comp.	method	lev.	cores	runtime
pilotCost	off	off	none	1	1	10.3k
pilotCost	off	off	none	2	3	10.4k
pilotCost	on	off	none	7	1	13.7k
diversity	on	on	zpaq	9	1	1.66M
diversity	off	on	zpaq	9	1	1.61M
diversity	off	on	lzma	9	4	288k

Rows are real selected configurations from the $n = 16$ samples. enc. and comp. are selected/deselected options; runtime is the cost paid before fitting the energy predictor.

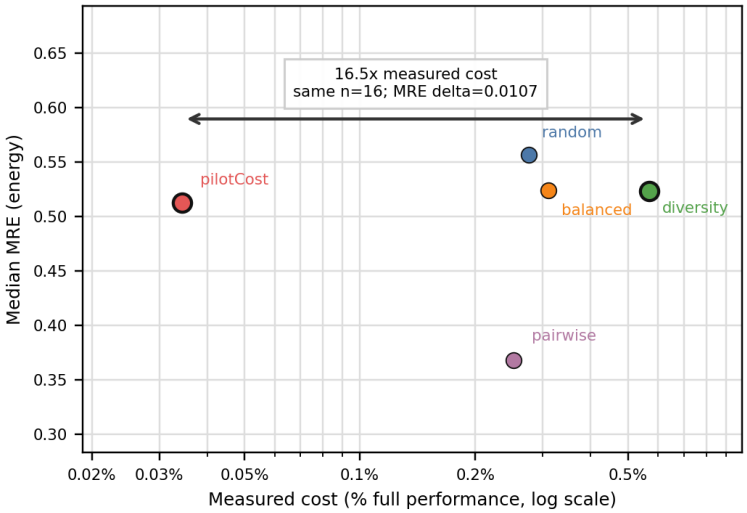


Fig. 1: Integrated running example on lrzip 0.631 compressing a 621 MB UIQ2 workload file (5,184 configurations, 12 option columns, runtime and energy measured over five repetitions). Left: the sampling principle and actual configuration excerpts show how two same-size samples can visit different option regions. Right: all strategies use $n = 16$, but measured performance cost separates nearly equal-error choices: pilot-cost and diversity differ by $16.5\times$ measured cost while MRE changes only from 51.3% to 52.4%.

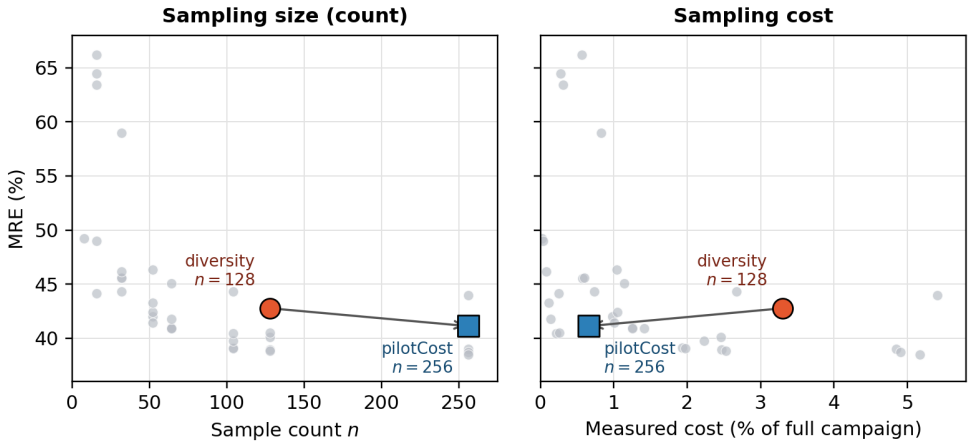


Fig. 2: A larger lrzip sample can be cheaper and more accurate under measured cost. The left panel uses sample count as the cost axis; the right panel uses measured acquisition cost for the same deployable strategy–budget combinations. The orange circle is `diversity` at $n = 128$ (MRE 42.77%, measured cost 3.297% of the full campaign). The blue square is `pilotCost` at $n = 256$ (MRE 41.11%, measured cost 0.658%). Grey points are the remaining deployable combinations in the same scenario.

- A quantitative decision-impact analysis: replacing count by measured cost changes 538/1,666 top strategy-budget decisions across six evidence families.
- An empirical characterization of when the proxy fails: heavy-tailed costs, target-cost asymmetry, workload shifts, and diversity/uncertainty sampling.
- An artifact enabling future studies to apply the same audit to their own data.

The remainder follows the audit structure. Section II reviews the sampling setting for practitioners. Section III defines measured cost and the budget ratio. Section IV states the research questions and the audit protocol. Section V reports the decision-impact results. Section VI draws practical implications for future studies. Section VII discusses validity, Section VIII places the work, and Section IX closes with a call to action.

II. Background

A configurable system exposes options whose combinations form a configuration space C. For a fixed workload, each configuration $c \in C$ has a measured value y_c—runtime, energy, output size—and an acquisition cost—the resource consumed to measure it. Sampling-based performance prediction trains a regression model $\hat{y}$ from a measured subset $S \subset C$ and evaluates error on held-out configurations T [1]–[4]. The standard effort axis in this literature is $n = |S|$, the number of measured configurations.

Sampling strategies.: Studies compare several sampling families. *Random* draws uniformly. *Solver-based* uses SAT

solvers for valid configurations; *randomized solver-based* perturbs the solver order [5]. *t-wise* covers option-value combinations [6]–[8]. *Distance-based* and *diversified distance-based* spread samples in feature space [2]. Pool-based analogues include *balanced* (keeps option frequencies), *coverage* (matches marginal coverage), *diversity* (maximizes Hamming distance), *uncertainty* (picks high-residual configurations), *pairwise* (covers option pairs), and *pilot-cost* (learns a cheap cost model first). Cheap/expensive oracles bound the diagnostic extremes.

How prior work counts cost.: Prior work equates measurement effort with sample size. Sarkar *et al.* collapse acquisition cost to a term proportional to n [9]; Westermann *et al.* count measurement points [10]; Siegmund *et al.* and Kaltenecker *et al.* compare strategies at common n [1], [2]. The premise that n is a valid cost proxy has not been tested. Guizzo *et al.* showed an analogous surrogate-cost threat in mutation testing [11]; we ask the same question for configurable-system sampling.

III. Cost Model

For a run r of strategy s at training size n, let $S_{r,s,n} \subseteq C$ be the selected sample. The usual cost model is the count of measured configurations:

$$SizeCost(s, n) = n. \tag{1}$$

We compare it with measured training acquisition cost:

$$TrainCost(r, s, n) = \sum_{c \in S_{r,s,n}} cost(c). \tag{2}$$

Here $cost(c)$ is the measured resource consumed to acquire the label for configuration c. In most of our study it is wall-clock measurement time; in the performance–energy family it can be energy; in the Linux family it is the uncompressed `vmlinux` size, representing a storage-bound cost. The model does not require the cost dimension to equal the prediction target: one may predict output size while paying time, or predict runtime while paying energy.

To compare systems with different absolute costs, we normalize by the cost of measuring the whole available configuration space:

$$FullCost = \sum_{c \in C} cost(c), \tag{3}$$

$$TrainCostPct = \frac{TrainCost}{FullCost}, \qquad SizePct = \frac{n}{|C|}. \tag{4}$$

The count proxy has good *budget agreement* when the fraction of configurations measured matches the fraction of measured cost spent. We summarize this with a budget ratio:

$$Ratio(r, s, n) = \frac{TrainCostPct(r, s, n)}{SizePct(s, n)}. \tag{5}$$

$Ratio = 1$ means a 10% sample consumes about 10% of the full measurement cost. $Ratio > 1$ means the strategy selected costlier-than-average configurations; $Ratio < 1$ means cheaper-than-average ones. $Ratio$ is the key diagnostic tool: it exposes whether a strategy systematically overspends or underspends its count budget.

For decision-level comparisons, we use two views. First, we rank strategy–budget combinations with a normalized objective:

$$J_\lambda^{\text{size}} = \widetilde{MRE} + \lambda \widetilde{SizeCost}, \tag{6}$$

$$J_\lambda^{\text{meas}} = \widetilde{MRE} + \lambda \widetilde{TrainCost}, \tag{7}$$

where the cost weight $\lambda \in \{0.1, 0.25, 0.5, 1, 2, 4, 10\}$ spans accuracy-heavy to cost-heavy choices. Second, we compare Pareto fronts in the MRE–cost plane: a combination is efficient if no other combination has both lower error and lower cost. The main threat we test is whether the recommended strategy or the efficient set changes when $SizeCost$ is replaced by $TrainCost$.

A strict dominance case is instructive: if the count-cost winner is both less accurate and more expensive under measured cost, the count recommendation is clearly wrong. More commonly the count winner is more accurate but costlier; this is a real trade-off. Our audit distinguishes both cases.

For practitioners, the three claim types are:

- **Fixed-size accuracy claim.** Which strategy predicts best when every strategy receives the same number of training configurations?
- **Count-budget claim.** Which strategy is preferable when sample count is the intended budget?
- **Measured-budget claim.** Which strategy is preferable when the resource actually paid for training labels is charged?

These can agree, and in several of our validation cases they do. They are nevertheless different claims. Reporting only the first while implying the third is the construct-validity gap this paper targets.

IV. Audit Framework and Study Design

The study asks one question: when is sample size a sufficiently good proxy for measured acquisition cost that a practitioner can trust it for decisions? We organize the evaluation around three research questions.

- **RQ1. To what extent does sample size reflect measured acquisition cost?** We test whether same-size samples spend comparable measured budgets, whether larger samples reliably cost more, and whether a fixed fraction of configurations corresponds to the same fraction of measured cost.
- **RQ2. Does replacing sample size by measured acquisition cost change sampling-strategy conclusions?** We compare the top accuracy–cost choices and the Pareto-efficient sets under count-based cost and measured cost.
- **RQ3. Under which conditions is sample size a safe or unsafe cost proxy?** We analyze how the answer depends on cost heterogeneity, strategy bias, target–cost asymmetry, and workload or dataset family.

The three-step audit.: The methodological contribution is a lightweight audit any practitioner can run on their own data:

Step 1: Inspect cost spread. Compute the coefficient of variation (CV) and p_{90}/p_{10} of per-configuration acquisition costs on the available measurement pool or an initial pilot. A CV near or above 1.0 or $p_{90}/p_{10} \geq 1.5$ signals sufficient heterogeneity to warrant further checking.

Step 2: Report per-strategy budget ratios. For each sampling strategy at each nominal budget, compute $Ratio$ from Equation (5). A strategy whose $Ratio$ is consistently above or below 1 has systematic cost bias. The family-level median $Ratio$ reveals whether the count budget is calibrated.

Step 3: Recompute rankings and Pareto fronts. Replace $SizeCost$ with $TrainCost$ in the accuracy–cost objective (Equation (1) vs. Equation (2)) and recompute the top recommendation and the Pareto-efficient set. If the top choice and the frontier are unchanged, the count proxy is validated for that setting. If not, the conclusion should be stated under measured cost.

The audit separates two uses. In a *retrospective* study, measured cost is an evaluation axis: it checks whether a published fixed-n conclusion transfers to a measured-budget claim. In a *prospective* campaign, cost information can guide sampling. The audit handles both roles.

A. Evidence Families

We keep six evidence families separate because they stress different validity conditions. Pooling them would give a larger number but a weaker explanation of when the sample-size proxy is reliable.

The REPLICATION-ICSE19 family is the most conservative test. It reuses the published strategy panel—random, solver-based, randomized solver-based, t-wise, distance-based, and diversified distance-based—on seven Kaltenecker-style systems. Only the cost axis changes from count to measured performance cost. If a conclusion changes there, it is not caused by a new sampler or benchmark. The REPLICATION-ICPE20 family uses the same logic on x264 with 17 inputs and two properties, testing whether the cost proxy is safer when target and cost coincide than when they differ.

ENERGY asks whether the issue is only about runtime or also appears when performance and energy can each be target or cost. WORKLOAD keeps systems fixed but changes workloads, testing whether cost-proxy validity is a property of the system or the system–workload pair. LCBENCH moves outside configurable-software performance and checks whether runtime learning curves in hyperparameter optimization show the same count-cost gap. LINUX stresses storage-bound cost at scale across seven Linux releases.

B. Protocol

For each tuple of system, workload, target, strategy, training size, and seed, we replay or generate the training sample, compute $TrainCost$, train the predictor on configuration options, and evaluate MRE on held-out configurations. Stochastic strategies use repeated seeds when available. We exclude measured performance, energy, size, and runtime columns from the feature matrix to avoid label leakage.

Artifact.: Our anonymized artifact contains the selected configurations, strategy identifiers, seeds, measured targets, measured acquisition costs, and scripts for recomputing $TrainCost$, $Ratio$, top-choice flips, and Pareto fronts. It supports reproducing all tables and figures and applying the same audit to new data.

V. Results

One number summarizes the decision impact: across 1,666 context–weight decisions, replacing sample count by measured acquisition cost changes the top strategy–budget combination in 538 cases (32%). Of these, 293 (54% of changed) are accuracy–cost trade-offs—the count winner is more accurate but costlier. The stricter 57 cases (11% of changed) involve strict dominance: the count winner is not on the measured-cost Pareto frontier, meaning another combination is no less accurate and no more expensive. Count Pareto fronts miss about 52% of combinations efficient under measured cost.

The evidence families fall into three regimes. REPLICATION-ICSE19 is a mixed bag: $5/7$ systems validate the count proxy, while lrzip flips in most cost weights. REPLICATION-ICPE20 is stable when target and cost coincide (time/time), but $21/119$ decisions change under asymmetry (time-cost/ size-target). ENERGY changes top recommendations in $24/28$ scenarios. WORKLOAD changes $72/93$ contexts. LCBENCH changes $33/35$ datasets. LINUX changes on all seven releases. The pattern is consistent: the audit catches failures where cost heterogeneity meets strategy bias.

A. RQ1: Sample Size Orders Cost, But Does Not Budget It

The conclusion is clear: sample size orders cost but does not budget it. In every family, larger samples are positively rank-correlated with measured cost—Spearman $\rho = 0.879$–0.943 in REPLICATION-ICSE19, 0.710–0.996 in ENERGY, near monotone in LINUX. The ordering signal is real; count-based plots are not arbitrary.

The failure is at the budget level. At fixed n, strategies visit different cost regions. In the lrzip running example, five strategies at $n = 16$ span a $16.5\times$ measured-cost range. Cost spread ranges from benign (LLVM $p_{90}/p_{10} = 1.21$) to heavy-tailed (BerkeleyDBC $p_{90}/p_{10} \approx 47$, jump3r CV 7.104, Linux CV 0.99–1.43, LCBench CV 0.595–2.247). The budget ratio $Ratio$ confirms the calibration gap: a nominal 10% sample can spend close to 10% of the budget, but BerkeleyDBC has median $Ratio = 0.797$, ENERGY reaches $Ratio = 0.168$, and workload contexts span 0.661–1.059.

The x264 revisit shows why aggregate calibration is not enough. Time budgets look calibrated on average, yet strategies differ: Henard-style samples tend cheaper, t-wise samples costlier. A family can pass the coarse ordering check and still fail the same-size, strategy-specific budget check that matters for decisions.

TABLE I: Dataset characteristics. Configurations are measured configurations available per context unless noted otherwise; NFPs are the measured non-functional properties used as targets or acquisition costs.

Family	Scope	Configurations	Options/features	NFPs used	Inputs/workloads
REPLICATION-ICSE19	Seven configurable systems from Kaltenecker *et al.* [2].	432–68,640 valid configurations per system.	11–54 options.	One performance-like target per system; the same measured value is charged as cost.	One context per system.
REPLICATION-ICPE20	x264 ICPE 2020 [3].	1,152 valid configurations.	16 binary options.	Encoding time and output size, combined into four cost–target scenarios.	17 input videos.
ENERGY	Seven systems from the performance–energy corpus.	180–6,840 measured configurations per system.	System-specific option vectors from the source artifacts.	Performance and energy, combined into four scenarios per system.	One workload per selected system.
WORKLOAD	Nine systems from the workload-impact ICSE 2023corpus.	191–6,764 configurations per system; 25,258 measurements overall.	System-specific option vectors from the source artifacts.	Wall-clock time, or oriented throughput for H2.	6–13 workloads per system, 93 contexts total.
LCBENCH	35 OpenML datasets from LCBench.	2,000 hyperparameter configurations per dataset.	7 hyperparameters.	Validation accuracy as target, total runtime as cost.	35 dataset contexts.
LINUX	Seven TuxKConfig Linux releases.	19,952–92,562 valid configurations per release.	494–497 retained kernel options.	Uncompressed `vmlinux` size as target and storage-cost proxy.	Seven releases, v4.13–v5.08.

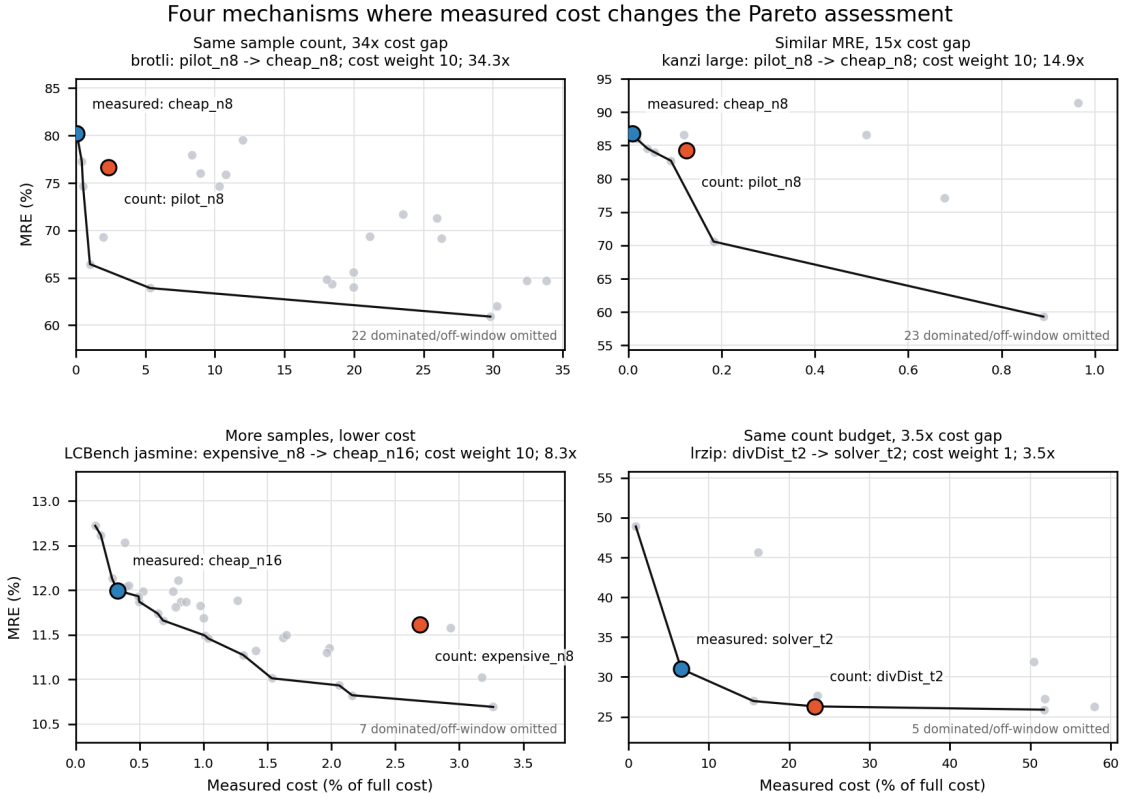


Fig. 3: Mechanism gallery for measured cost. Each panel plots the measured-cost Pareto space for one high-contrast case; lower-left is better. Orange marks the count-cost winner for the stated cost weight, blue marks the measured-cost winner, and the black polyline connects measured-cost Pareto-efficient combinations.

Figure 2 shows the sharpest failure. Under count cost, `pilotCost` at $n = 256$ is simply "larger" than `diversity` at $n = 128$. Under measured cost, `pilotCost` spends 0.658% of the campaign and `diversity` spends 3.297%—the larger sample is $5\times$ cheaper and slightly more accurate.

Answer to RQ1

Sample size is a useful ordering variable, but it is not a calibrated budget. Across all families, larger samples usually cost more; nevertheless, same-size samples can differ by orders of magnitude in measured cost, and a larger sample can be cheaper than a smaller one.

B. RQ2: The Audit Changes Decisions

Replacing count by measured cost changes $538/1{,}666$ decisions (32%). 57 of these (11% of changed) are strict dominance failures. Pareto frontiers miss 52% of measured-cost-efficient combinations.

Family by family:

REPLICATION-ICSE19. LLVM, Polly, Dune, and HIPAcc validate the count proxy. lrzip flips in $5/7$ cost-weight settings; frontiers change on $5/7$ systems. The replication shows the audit can validate and flag within the same family.

REPLICATION-ICPE20. The symmetric time/time case is stable—count and measured cost agree when the acquisi-

tion resource is the prediction target. The asymmetric time-cost/size-target scenario changes $21/119$ decisions and has median Pareto Jaccard 0.375. The cost dimension matters independently of the prediction target.

ENERGY. Top combinations change in $24/28$ scenarios. Pareto fronts differ in all 28. Target–cost asymmetry is the dominant mechanism: When target equals acquisition cost, recommendations are stable for $7/7$ weights; when target is energy but cost is time, they change for $6/7$ weights.

WORKLOAD. $72/93$ contexts change top recommendations; $71/93$ have different Pareto fronts. Workload shifts alter the cost distribution enough to flip decisions even when the system is fixed.

LCBENCH. $33/35$ datasets change recommendations; $32/35$ change Pareto fronts. The proxy failure extends beyond configurable-software systems to general hyperparameter-optimization learning curves.

LINUX. Every release changes top recommendations and Pareto fronts. Diversity and uncertainty sampling drive the gap: they select configurations with 1.24–$1.52\times$ the count budget share, while random-like strategies stay near $Ratio = 1$.

Answer to RQ2

Replacing sample size by measured cost changes both top recommendations and the efficient choice set. Across 1,666 context–weight decisions, the top strategy–budget combination changes in 538 cases (32%). The strict failures are the 57 changed cases where the count-cost winner is dominated under measured cost.

C. RQ3: When Does the Proxy Fail?

The count proxy is most defensible when per-configuration costs are tight, the target and acquisition resource coincide, and strategies are cost-blind. LLVM, Polly, Dune, HIPAcc, PostgreSQL, and x264 time/time show that count-based conclusions survive under these conditions.

It is most fragile when three conditions coincide: (1) heavy-tailed cost distributions, (2) target–cost asymmetry, and (3) diversity- or uncertainty-seeking strategies. Each condition alone can create a gap; all three together are the high-risk regime.

Figure 3 illustrates four recurring mechanisms: same-size cost gaps, similar-MRE/different-cost pairs, larger-but-cheaper samples, and strategy separation under measured cost. These are not exhaustive but representative of how the proxy fails.

The families fall into three practical regimes:

- **Validated transfer.** Count-based conclusions survive the audit. Examples: LLVM, Polly, Dune, HIPAcc, PostgreSQL, x264 time/time.
- **Warning.** Top choices survive but the efficient set changes. Examples: 7Z, BerkeleyDBC. A reader who wants the single winner may reach the same conclusion, but a reader comparing trade-offs would miss efficient alternatives.
- **Changed assessment.** Measured cost changes the recommendation itself. Examples: lrzip, asymmetric x264, heavy-tailed energy systems, most workload contexts, most LCBench datasets, every Linux release.

Answer to RQ3

Sample size is safest as a cost proxy when per-configuration costs are tight, the target and bottleneck resource coincide, and strategies are cost-blind. It is unsafe when costs are heavy-tailed, target and acquisition cost differ, workload or dataset changes the cost distribution, or the strategy deliberately seeks diverse or uncertain regions.

VI. Discussion

What should prior count-based results mean?: The audit does not make prior studies "wrong." It shows that a size-only cost claim is under-specified. When a study compares strategies at the same n and claims only prediction accuracy, our evidence does not invalidate that claim. The problem starts when the same result is read as an acquisition-cost claim, or when a larger n is assumed to mean a larger real budget. The safest interpretation is conditional: count-based conclusions transfer when costs are homogeneous enough or strategies are cost-blind enough; they need re-evaluation when strategies visit systematically cheaper or costlier regions.

Three claim types.: Every sampling study makes one of three claims. A *fixed-size accuracy claim* says which strategy predicts best at the same n. A *count-budget claim* says which strategy is preferable when sample count is the budget. A *measured-budget claim* says which is preferable under the resource actually paid. These can agree, and in several validated cases they do. They are nevertheless different claims. Reporting the first while implying the third is the gap this paper targets. The audit converts a fixed-size claim into a measured-budget claim in three steps, and any paper that makes a resource claim should run those steps.

What should future papers report?: The reporting change is lightweight. Every sampling paper should:

1) State the cost dimension and report per-configuration cost spread (CV and p_{90}/p_{10}).
2) Report $Ratio$ by strategy at the nominal budgets used. A global average can hide a strategy that is consistently cheap or expensive.
3) If the paper recommends a strategy, recompute the ranking and Pareto front with measured cost. If the front and top choice are unchanged, the count proxy is validated. If not, state which conclusion is count-based and which is measured-cost based.

Artifacts should record selected configurations, strategy identifiers, seeds, measured targets, and measured acquisition costs. These fields are enough to recompute rankings and Pareto fronts.

A practical incremental path.: In a prospective campaign, cost information can guide sampling before all measurements are taken. A practical path: start with a small pilot, estimate whether costs are homogeneous enough to keep using count, and only consider a cost-aware sampler if the pilot shows heavy tails or strategy-specific cost bias. This keeps the

recommendation modest. The paper does not require every study to optimize acquisition cost; it requires studies that make resource claims to validate the resource proxy they use.

When does measured cost matter most?: The clearest warning signs are heavy-tailed costs (CV ≥ 1.0, $p_{90}/p_{10} \geq 1.5$), target–cost asymmetry (predicting a different resource than paid for acquisition), and strategy bias ($Ratio$ consistently above or below 1 for a strategy). Any one of these justifies running Step 3 of the audit. All three together guarantee that conclusions will change.

VII. Threats to Validity

Cost construct.: We define acquisition cost as the sum of measured per-configuration costs in existing artifacts. This captures the benchmarked resource, not production overhead such as orchestration or warm-up. The relative questions we ask—whether same-size samples choose cheaper or costlier configurations—are less sensitive to machine-specific absolute times.

Protocol heterogeneity.: The evidence families use different strategy panels, learners, workloads, and cost dimensions, so we report them separately rather than pooling one effect size. RQ1 is learner-independent; RQ2 and RQ3 depend on MRE and could move with another learner or error metric. We keep original learners for the two replication families and report learner-sensitivity checks for newly framed families.

Learner choice.: A better learner can improve MRE and change the winning strategy–budget combination. It does not define acquisition cost: RQ1 is computed before fitting any model. The Linux HGBT sensitivity run improves accuracy substantially, yet measured cost still changes rankings and Pareto fronts on every release. Better predictors can change which combination wins; they do not make sample size a sufficient cost model.

External validity.: The systems cover compilers, encoders, databases, archivers, energy measurements, workloads, ML runtime, and Linux releases, but they remain benchmark artifacts. Practitioners should transfer the check, not the exact thresholds: inspect cost spread and per-strategy budget ratios on the target campaign.

VIII. Related Work

Sampling configurable systems.: Sampling-based performance prediction evaluates random, solver-based, coverage-based, distance-based, and transfer-learning strategies [1], [2], [4], [9], [12]. Effort is overwhelmingly measured as sample size. Our contribution is not a new sampler; it is a validity study of that effort axis.

Surrogate and heterogeneous costs.: Replacing a real resource with an easy-to-count surrogate is a recurring validity concern. Guizzo *et al.* documented it in mutation testing, where mutant count can mislead relative to execution time [11]. We ask the analogous question for configuration sampling. The mechanism is specific to configurable systems: sample size is correlated with measured cost, yet strategy-specific budget ratios and decision fronts can still change because acquisition cost varies widely when configurations are slow, energy-intensive, storage-heavy, or workload-dependent.

Counting effort versus measuring acquisition cost.: This is the gap we target. Across tuning, transfer, and prediction, "effort" is overwhelmingly expressed as counts—number of measurements, evaluations, or queries [13]–[18]. Counts order effort, but they are a budget only if one count unit stands for a fixed amount of acquisition cost. When it does not, a "cost-effective" result is an accuracy result wearing a cost label. Our decision tests show exactly where this matters: when measured cost diverges from the count, the recommended strategy or the non-dominated set can change. Part of the literature already grounds counts in measured budgets [19], [20], which is exactly our recommendation. Our results push one step further: even case-level calibration can drift when measurement costs are heterogeneous within a system or biased by a strategy.

IX. Conclusion

The practical fix is a lightweight three-step audit: inspect cost spread, report per-strategy budget ratios, and recompute rankings and Pareto fronts under measured cost. Running this audit across six evidence families and $1{,}666$ decisions reveals that sample size changes the recommended strategy–budget combination in 538 cases (32%), and 57 of the count winners are strictly dominated. Count-based Pareto fronts miss about 52% of measured-cost-efficient combinations.

The proxy is safest on homogeneous systems with cost-blind samplers and target–cost symmetry. It is unsafe under heavy-tailed costs, target–cost asymmetry, workload shifts, and diversity/uncertainty sampling. These conditions are common, measurable, and severe enough to change conclusions.

The burden is on future studies to show that their count budget is decision-equivalent to the measured resource, or to state the conclusion under measured cost. Our artifact makes running this audit a matter of minutes, not months.

Acknowledgments

Anonymized for review.

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